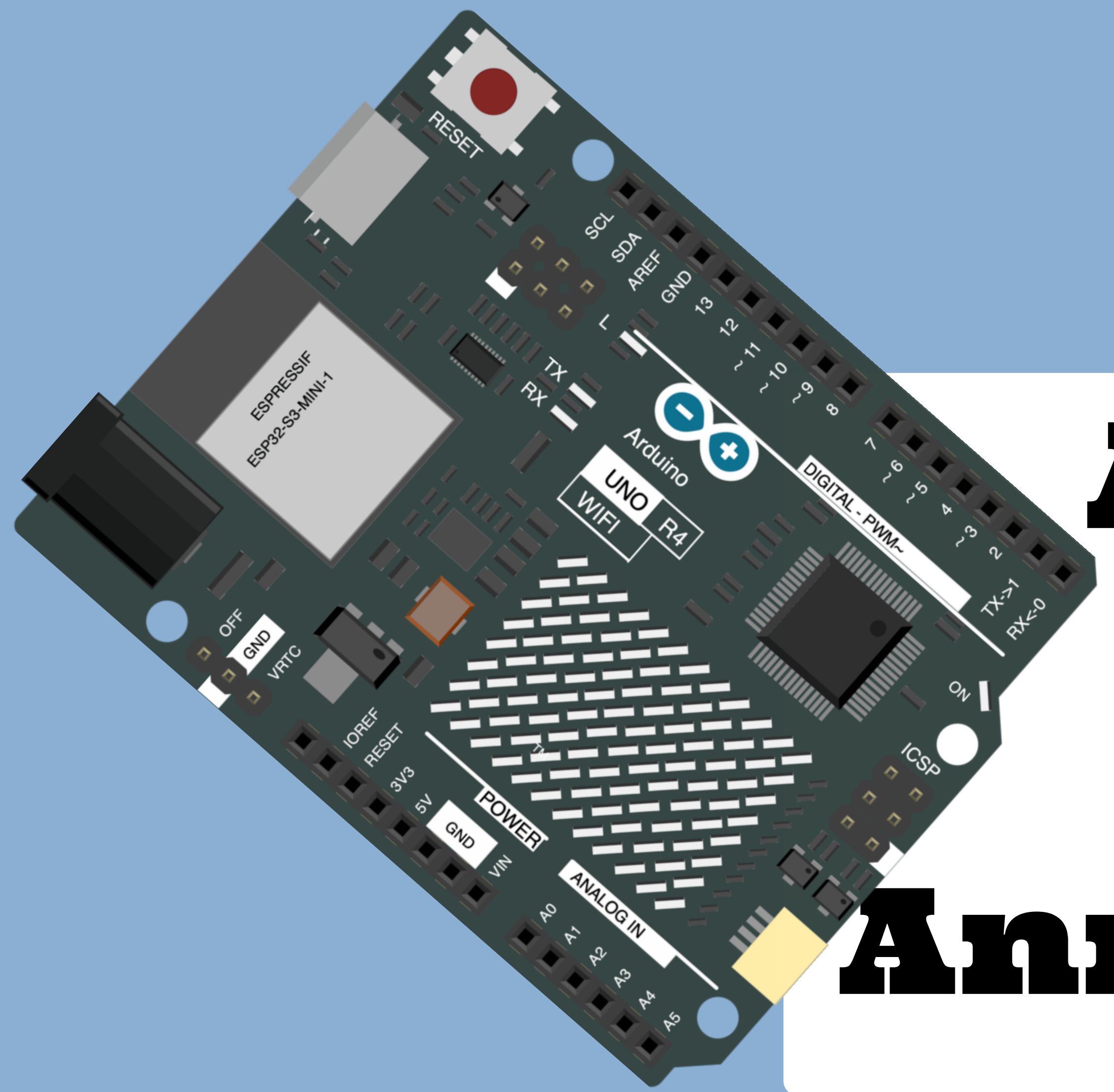
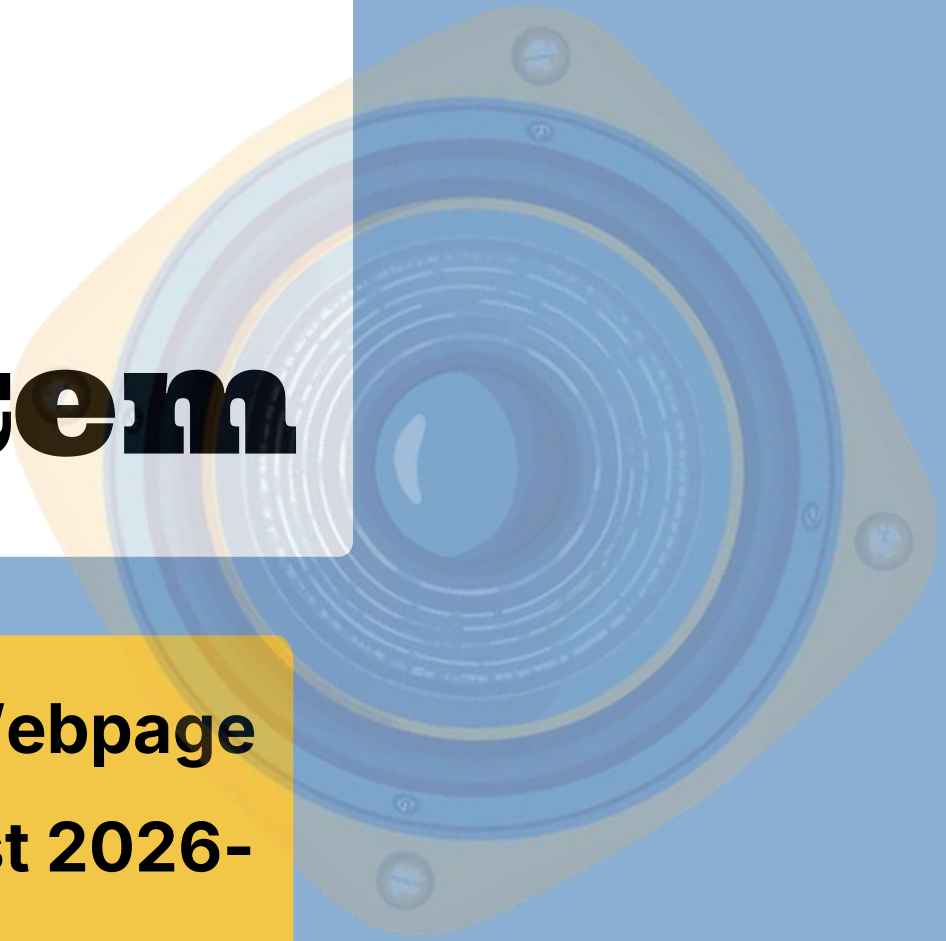
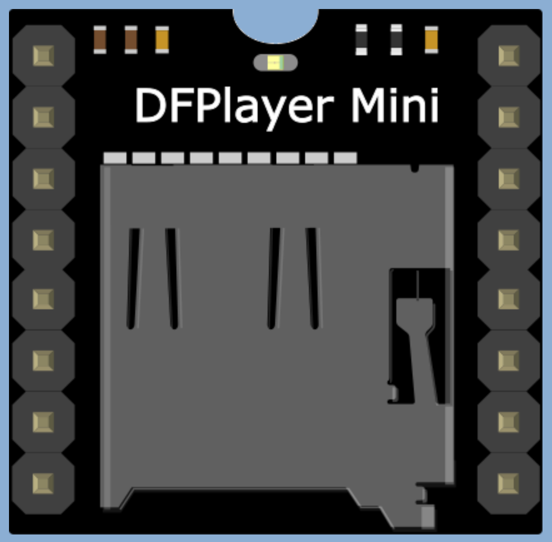


Automatic School Buzzer and Announcement System



Using Arduino Uno R4,DF Player Mini, Speaker, and Webpage

School Name: Abhinav Vidya Mandir, VIDYA Tech Fest 2026-27



Problem Statement

Manual bell ringing causes delays and disruptions.

Announcements need constant staff presence.

Schools urgently need automation solutions.

Time and effort are wasted on manual tasks.

By the end of this project, you'll be able to...

🔔 Ring the school bell automatically

📢 Play recorded announcements

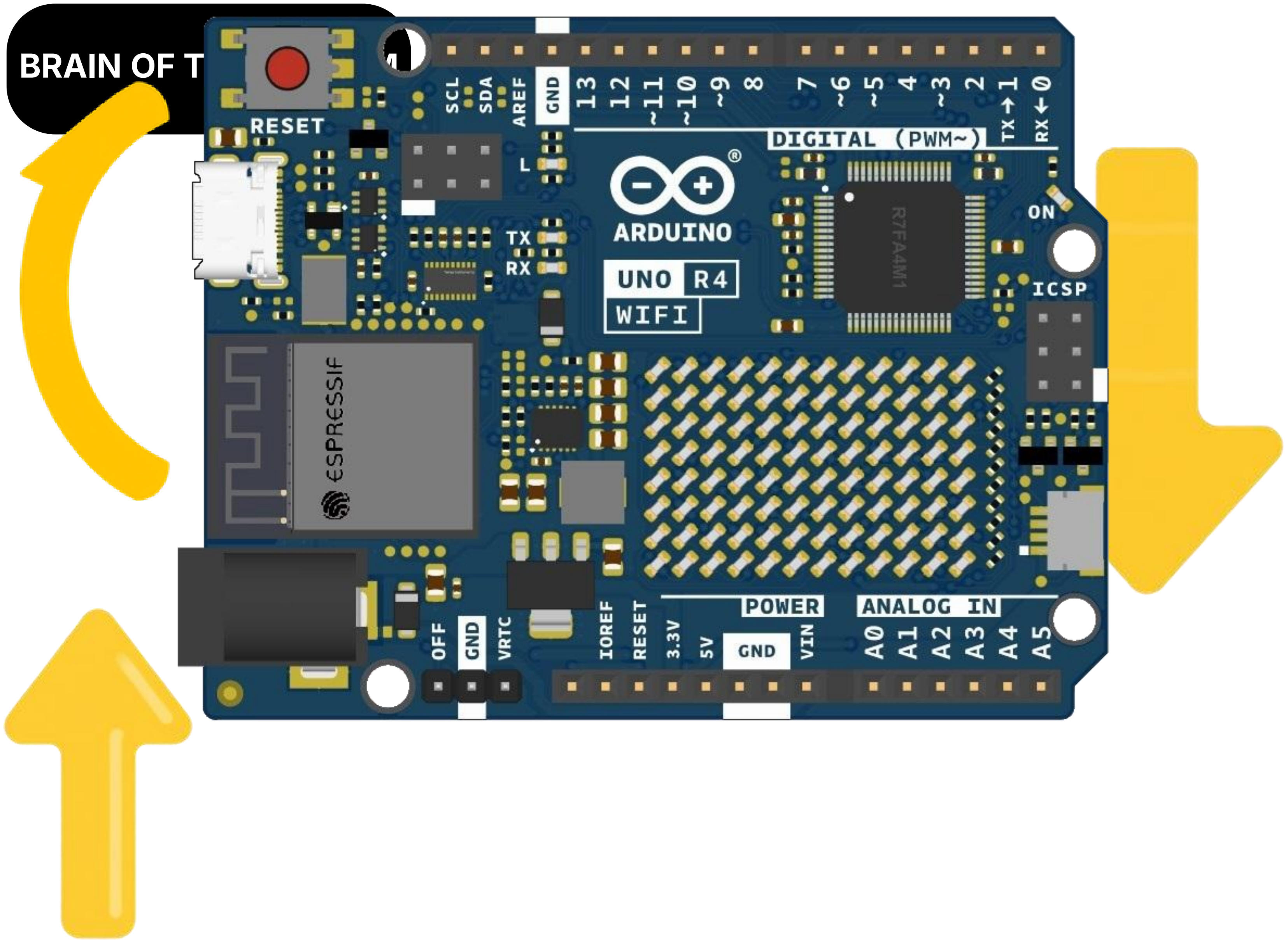
🕐 Show current time on Dashboard Webpage

✅ Reduce manual work in schools

Introduction to Arduino

Arduino R4 is a microcontroller board that controls electronic devices and components.

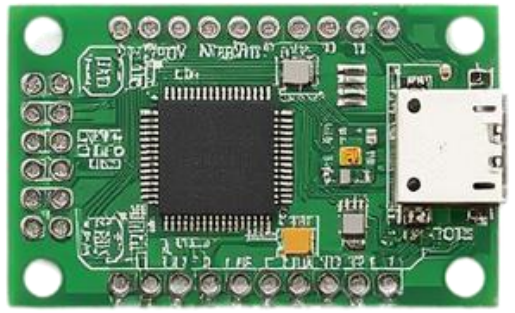
It is easy to use in school projects and is the main controller of this system.



Components Required

Below are the key components used in the Automatic School Buzzer and Announcement System:

Arduino Uno



RTC DS3231

DF Player Mini

Speaker

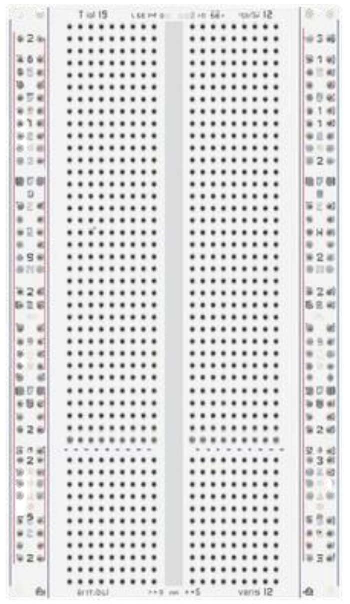
LCD 16x2

Buzzer

Components Required (cont.)

Additional parts needed to build the circuit.

Jumper Wires
Connect components on the breadboard easily.



Breadboard
Prototype circuits without soldering.

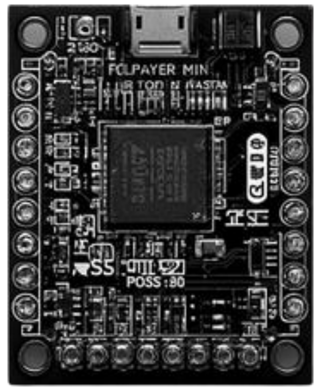
Power Supply
Provides stable power to Arduino and modules.

Micro SD Card
Stores audio announcement files for DF Player Mini.

Component Functions

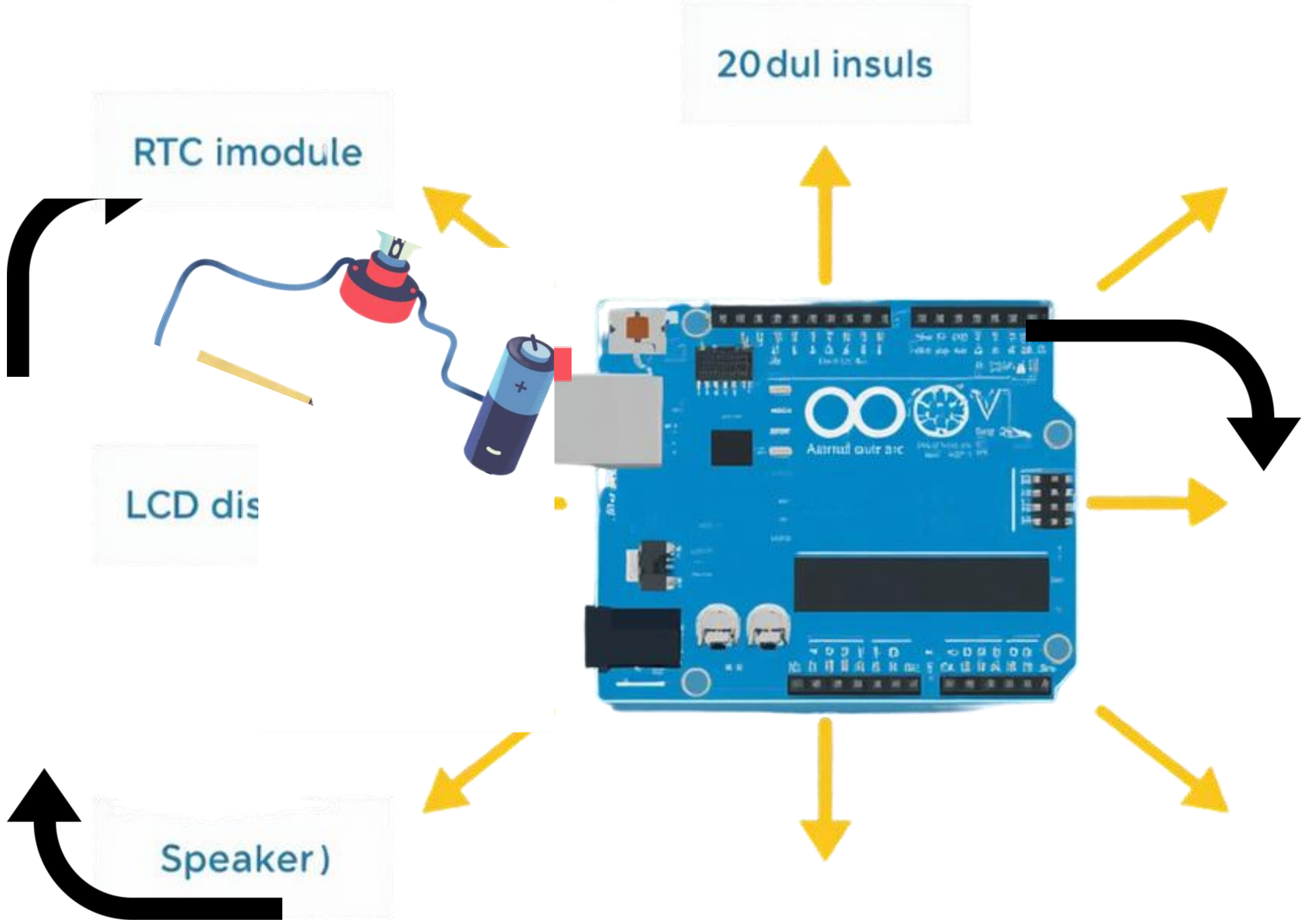
Each component has a specific role in the Automatic School Buzzer and Announcement System.

ARDUINO UNO – CONTROLS THE SYSTEM
RTC MODULE – KEEPS ACCURATE TIME
DF PLAYER MINI – PLAYS AUDIO FILES



SPEAKER – OUTPUTS ANNOUNCEMENTS
LCD DISPLAY – SHOWS TIME & STATUS
BUZZER – RINGS THE SCHOOL BELL

Block Diagram



RTC DS3231 → Arduino Uno → LCD Display

(Time input to controller to screen output)

Arduino Uno → DF Player Mini → Speaker

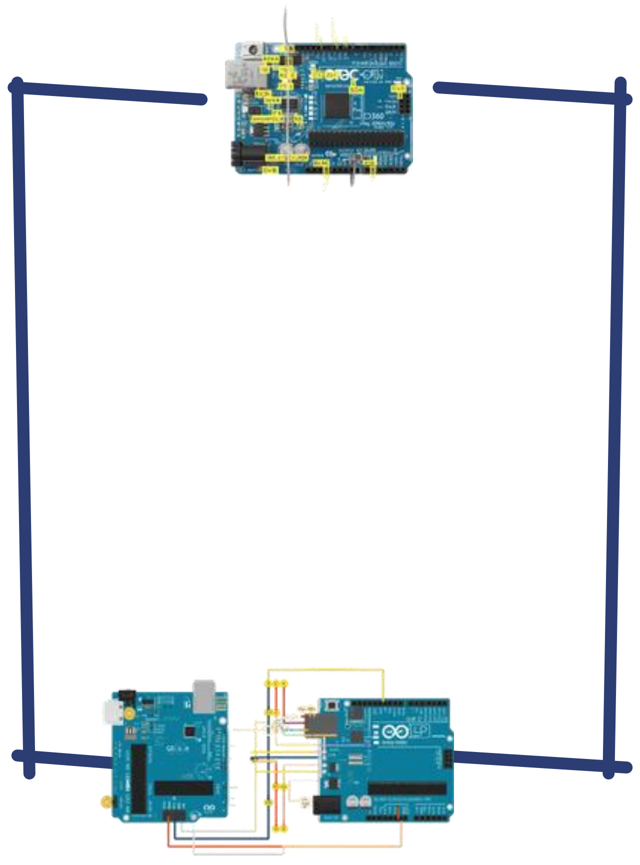
(Controller triggers audio playback)

Arduino Uno → Buzzer

(Controller rings the bell at scheduled times)

Circuit Connection Diagram

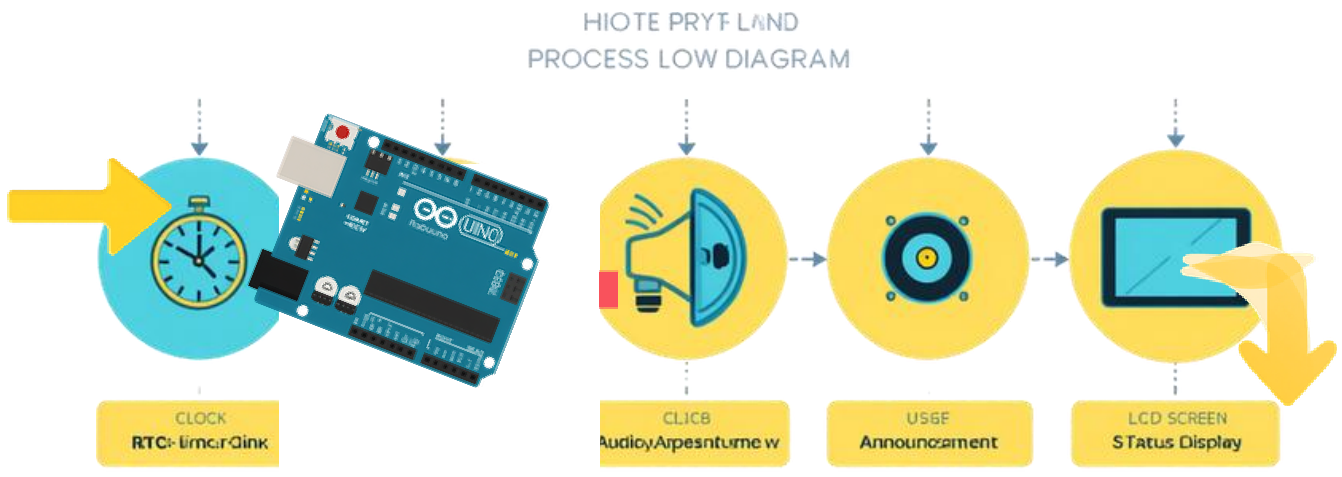
WIRING GUIDE



RTC & LCD
to Arduino

DF Player Mini,
Speaker & Buzzer

Working Principle



Grade 6-10

Step 1: RTC checks the current time and sends it to Arduino continuously.

Step 2: Arduino compares current time with the stored bell schedule.

Step 3: Buzzer rings at bell time.

Step 4: DF Player Mini plays the announcement.

Step 5: LCD shows the status message.

Audio Announcement Feature

Morning Assembly Announcement

Recess Bell Announcement

School Closing Announcement

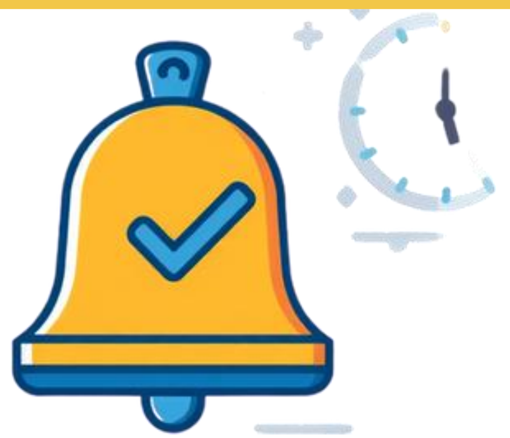
Emergency Message Playback

Advantages

Key benefits of the Automatic School Buzzer and Announcement System:

Fully Automatic

Saves Teacher's Time



Accurate Bell Timing

Clear Announcements

Easy to Modify

Low Cost

Real-Life Applications & Future Improvements

Applications

✓ Schools

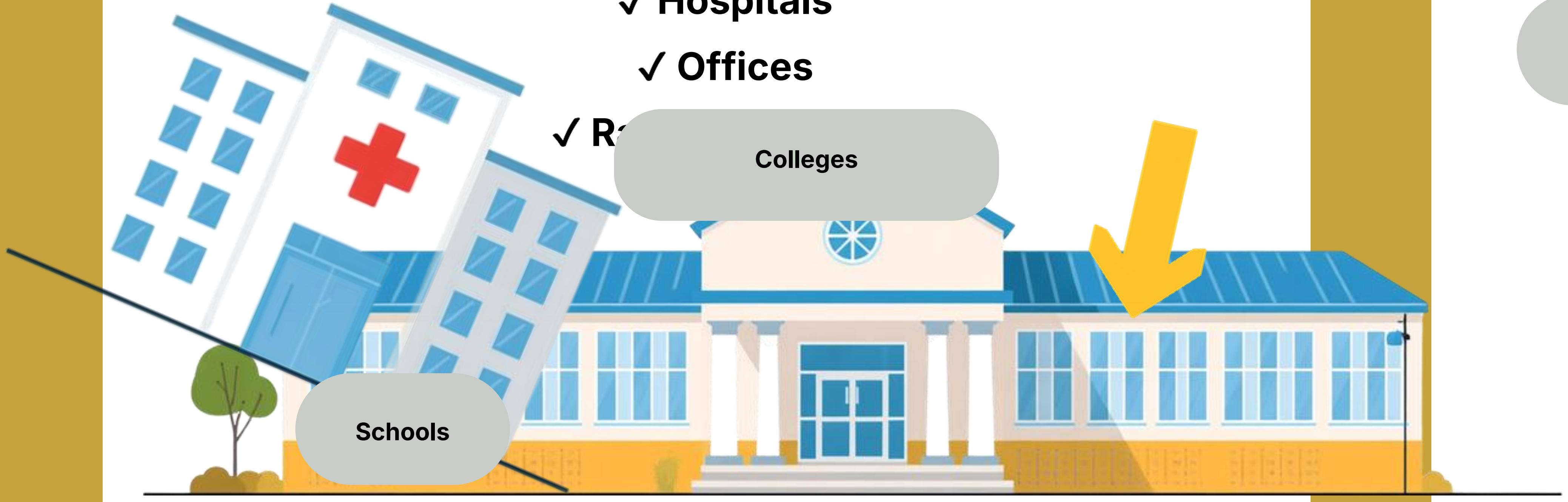
✓ Colleges

✓ Factories

✓ Hospitals

✓ Offices

✓ Ra



Future Ideas

- Mobile App Control
- Wi-Fi Monitoring
- Multiple Bell Schedules
- Voice Recording
- Cloud-Based Manag

Thank You!